A physically consistent representation of recurring Jökulhlaup floods in a fully-coupled subglacial hydrology model

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ABSTRACT. The design for the *Journal of Glaciology* has been implemented as a LaTeX 2ε class file and is derived from article.cls. We recommend that

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1 INTRODUCTION

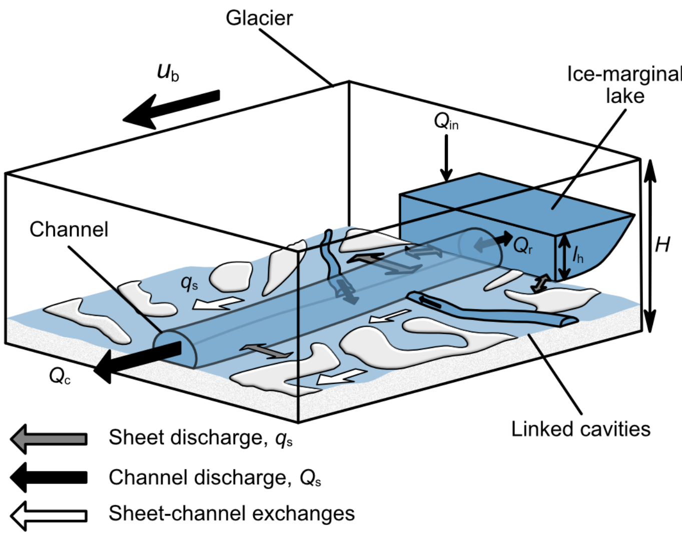
1. Importance of ice-marginal lake drainage and potential hazard implications
2. Efforts to model
3. shortcomings and opportunities to improve

2 METHODS

In this paper we build upon earlier 1D work (Kingslake and Ng, 2013) by incorporating a physically-consistent representation of ice-marginal lake drainage into the 2D Glacier Drainage System (GlaDS) model (Werder and others, 2013), part of the Ice-sheet and Sea-level System Model (ISSM; Larour and others, 2012, , `commit a9a3804`). Our work includes multiple couplings between the subglacial hydrological system and the ice-marginal lake, as well as between each of these and ice-velocity (Fig. 1). In Section 2.1 we describe the coupled ISSM-GlaDS model and relevant additional modifications to the version originally presented by (Werder and others, 2013). We describe the equations governing ice-marginal lake drainage in Section 2.2, and our method of numerical solution is given in Section 2.3.

2.1 Ice Flow and subglacial drainage models

We used the GlaDS model as integrated into ISSM, a finite element higher-order ice-flow model which operates on anisotropic meshes and can be fully-coupled to additional modelling components (Larour and others, 2012). GlaDS itself is a widely used (REF) 2D continuum model of subglacial hydrology. We refer interested readers to Appendix A.1 for an overview of key GlaDS equations, and to Werder and others (2013) for a full description of the equations governing water flow. In summary, GlaDS operates on an



[!h]

Fig. 1. Schematic illustration of our model, showing an ice-marginal lake connected two-ways to a subglacial channel and sheet. This in turn modifies basal velocity, u_b through effective pressure, N. Note, although the ice-marginal lake is shown as connected to one channel, in our model an ice-marginal lake can be connected to the subglacial system at multiple outlets. `[SINGLE-COLUMN]`

anisotropic mesh generated using ISSM and includes a description of i) distributed flow through a series of linked cavities represented as a continuous sheet with variable thickness, and ii) channelised flow through Röthlisberger channels that are represented along element edges. Sheet elements exchange water with channels, and the cross-sectional area of these channels evolves through time due to the dissipation of potential energy, sensible heat exchange, and cavity closure rates due to viscous ice creep. Discharge in both the distributed (q_s) and channelised system (Q_c) is driven by gradients in the hydraulic potential, $\nabla\phi$, with steeper gradients driving enhanced water discharge. In GlaDS, Q_c is always non-zero along every internal edge, though only a few reach a discernible discharge. For ease of visualisation we follow previous work [REF] in declaring edges where $Q_\mathrm{c} \geqslant 0.5$ m^{3}s^{-1} to be 'meaningful channels' (e.g., Figure 2c–e).

In addition to the ice-marginal lake modifications which we describe below (Section 2.2), we note two key differences between the original GlaDS (Werder and others, 2013) and the ISSM version used here: i) rather than assuming water flow within the sheet to always be turbulent (cf: Werder and others, 2013), we make use of the Hill and others (2024) transition parametrisation, whereby sheet flow switches between laminar and turbulent flow according to the local Reynolds number, and ii) following Stevens and others (2022), we implemented a viscously deforming 'elastic sheet layer' which crudely represents hydraulic jacking where water pressure exceeds ice overburden pressure (see Section A.1).

Finally, to capture the influence of ice-marginal lake drainage of ice dynamics, we include a two-way coupling between ice flow and subglacial hydrology. The GlaDS derived effective pressure ($N = p_{\mathrm{i}} - p_{\mathrm{w}}$, where p_{i} is ice pressure and p_{w} is water pressure) modifies basal velocity u_{b} by governing frictional resistance at the ice-bed interface. In turn, the magnitude of u_{b} modifies the rate at which cavities open and thereby controls the thickness of the sheet (Eq. A7–A8*). Two-way coupling between ISSM and GlaDS is a relatively new development, and the influence of friction parameters remains the subject of ongoing research (e.g., Khan and others, 2024), though model response is known to be strongly influenced by the choice of friction law (Choi and others, 2023). We view this work as necessarily explorative [Flesh this out more], and we do not undertake a full analysis of the modelling interactions between subglacial hydrology and ice dynamics, instead leaving this to others. Rather, we seek to capture the broad influence ice-marginal lake drainage cycles have upon ice dynamics. Accordingly, we present results using two widely applied friction laws: i) the 'Budd-style' friction law [REF] [MOVE EQUATIONS TO APPENDIX] implemented in ISSM in terms of basal stress with the form:

$$\tau_{\mathrm{b}} = C_{\mathrm{b}}^2 N^{\mathrm{r}} u_{\mathrm{b}}^s, \tag{1}$$

where τ_{b} is the basal stress magnitude, C_{b} is the friction coefficient, N is effective pressure, u_{b} is basal velocity magnitude, and both r and s are friction coefficients, or ii) a Schoof-style friction law [REF] with the form:

$$\tau_{\mathrm{b}} = \frac{C_{\mathrm{s}}^2 v_{\mathrm{b}}^m}{\left(1 + \left(\frac{C_{\mathrm{s}}^2}{C_{\max} N}\right)^{1/m} u_{\mathrm{b}}\right)^m}, \tag{2}$$

where C_{s} is the *Schoof* friction coefficient, m is a friction law exponent, and $C_{\max}$ is Iken's bound.

2.2 Lake evolution

Following Kingslake and Ng (2013) lake depth, l_{h} (Fig. 1), evolves according to the continuity equation:

$$\frac{\mathrm{d}l_{\mathrm{h}}}{\mathrm{d}t} = \frac{1}{A_{\mathrm{L}}}[Q_{\mathrm{in}} - Q_{\mathrm{r}}(t-1)], \tag{3}$$

where A_L is the uniform lake area, Q_{in} represents extraglacial lake input (e.g., rainfall, surface runoff) and Q_{r} represents inflow/outflow at the lake outlet(s) from the previous timestep. In moving from a 1D to 2D representation of subglacial hydrology and including the contribution from the sheet, Q_{r} in our model is given by:

$$Q_{\mathrm{r}}(t) = \sum_{i=1}^{N} \left(q_{\mathrm{s},i}(t) + Q_{\mathrm{c},i}(t) \right), \tag{4}$$

or the sum of the width integrated sheet discharge, q_{s}, and channel discharge, Q_{c} along N vertices connected to an ice-marginal lake. In GlaDS, a specific hydraulic potential is prescribed at outlets (Dirichlet boundary conditions), which is typically equivalent to zero water pressure (i.e., atmospheric pressure Werder and others, 2013). As a boundary condition at lake outlets, hydrostatic lake pressure fixes the lake outlet hydraulic potential, ϕ_{outlet} to be:

$$\phi_{\mathrm{outlet}}(t) = \rho_{\mathrm{w}} g z_{\mathrm{b}} + \rho_{\mathrm{w}} g l_{\mathrm{h}}(t), \tag{5}$$

at all times. By changing the hydraulic potential around the lake outlet, and in turn sheet/channel discharge (Eq. A4, & A11), this condition handles the coupling between subglacial hydrology and the ice-marginal lake.

2.3 Method of numerical solution

We solve for l_{h} using a Picard fixed-point iteration scheme comprising an outer and inner loop. In the outer loop, we update l_{h} given Q_{r} from the previous timestep (Eqn. 3) using a backward Euler solver and use this to set ϕ_{outlet} (Eqn. 5) at the current timestep. In the inner loop, GlaDS solves for the domain-wide ϕ until it satisfies a set convergence criteria, as in previous implementations of GlaDS (see: Werder and others, 2013; Hill and others, 2024). Following the inner loop converging, and given the global ϕ, the model updates h_{s}, N,Q_{c}, and q_{s}, before summing Q_{c} and q_{s} at lake outlet nodes to give Q_{r} (Eqn. 4). The outer loop iterates until the norm of the current and previous l_{h} satisfies a given tolerance.

3 MODEL APPLICATION TO A SYNTHETIC DOMAIN

We test our model on a simple 10×3 km rectangular domain, discretised on a 2D triangular mesh with a characteristic edge length of 100 m (n_{vertices}=2486). The bed elevation, h_{b} is characterised by a gentle slope (sin = 0.0075) dipping towards the terminus, and ice thickness is parabolic ($H \propto \sqrt{x}$) increasing from from H=50 m at X= 0 km and H = ∼155 m at X = 10 km. To facilitate comparison with previous work, our domain is similar in scale and geometry to that in Kingslake and Ng (2013), chosen to be loosely

Table 1. Variables and Units[a]

Variable	Units	Description
ϕ	Pa	Hydraulic potential
$Q_{\rm c}$	$\rm m^3 s^{-1}$	Channel discharge
$q_{\rm s}$	$\rm m^2\, s^{-1}$	Sheet discharge
$h_{\rm s}$	m	Sheet thickness[b]
$l_{\rm h}$	m	Lake Height
$Q_{\rm r}$	$\rm m^3 s^{-1}$	Lake discharge
u_b	$(m\, a)^{-1}$	Basal velocity
S	$\rm m^2$	Channel cross-sectional area
t	yrs	Time coordinate
X	km	Horizontal coordinate
Y	km	Vertical coordinate
N	Pa	Effective pressure

[a]The first section lists dependent variables, the second section lists coordinates, and the third section lists derived variables [FIX ORDER.]

[b]Note this includes the elastic sheet and cavity sheet thickness, see Eq. A6–A7

analogous to the Merzbacher Lake and South Inylchek Glacier, Kyrgyztan (Ng and Liu, 2009).

Model variables are given in Table 1 and our synthetic model parameters are given in Table 2. The parameter dependency of GlaDS has already been extensively explore elsewhere [REF]. To efficiently explore the parameter dependence of our ice-marginal lake drainage model we limited our testing to those previously highlighted as critical to subglacial drainage (e.g., channel/sheet conductivity terms, $k_\mathrm{c}/k_\mathrm{s}$, the basal bump height, h_b, the channel sheet width, l_c, and the cavity spacing, l_r) as well as the terms which control the input of water into both the subglacial system (basal melt rate), and the ice-marginal lake (lake refill rate, Q_in). As is common-practice [REF], we varied one parameter at a time between a minimum and maximum bound with the limits (Table 2) informed both by previous work [REF] and the limits of numerical stability. As reference, an additional 'default' simulation was taken as the approximate midpoint of each of these parameters (Fig. 2). Each simulation was run for a total of 15 years and had a 30 minute timestep. Simulations were initialised with u_b 100 m a^{-1}, ϕ equivalent to 90% of ice overburden pressure, and an initial sheet thickness, h_s equal to half the bump height, h_b. We prescribe ϕ terminus X =0 km to be $\approx 1\times10^4$ as our Dirichlet boundary condition, though arbitrary, this does not qualitatively affect our results below. We impose zero inflow as our Neumann boundary condition at the other three boundaries. The ice-marginal lake, connected to a single vertices where $(X, Y) = (10,\ 1.5)$ km, had an initial height of 30 m (l_h at t=0). The evolution of the reference case through time is shown in Fig. 2, and the sensitivity of lake height (l_h), flood discharge (Q_r), and basal velocity (u_b) through time is shown in Figs. 3 & 4).

3.1 Stable flood cycles

In our reference case, the ice-marginal lake undergoes recurrent, stable, and asymmetric lake filling and draining cycles (Fig. 2a). Likewise, each lake filling and draining cycle is associated with a recurring flood with stable maximum amplitude and duration (Fig. 2b). The lake only partially drains each flood cycle, and never reaches a height necessary to sustain flotation (N=0, Fig. 2f). For the initial 5 years of the simulation, as the model adjusts to initial conditions, peak flood discharge increases with each successive flood, and the lake reaches successively lower post-flood lowstands (Fig. 2a,b). Thereafter, the flood cycles reach a stable periodicity (Fig. 2f) and clockwise hysteresis with flood cycles occurring approximately every 1.5 years.

The evolution of the subglacial channel system through a single flood cycle is shown in Fig. 2c–e (an animated version is shown in SUPP MOVIE), and the width-averaged basal velocity response over the same

Table 2. [NOTE, this fits in two-column mode!] Input parameters in our synthetic domain. The parameters we varied, and the range over which they varied are highlighted in bold.

[!h]

Description	Symbol	Value(s)	Units
Gravitational acceleration	g	9.81	m s^{-1}
Latent heat	L	3.34×10^5	J kg^{-1}
Ice density	ρ_i	917*	kg m^{-3}
Freshwater density	ρ_w	1000	kg m^{-3}
Pressure melt coefficient	c_t	7.5×10^{-8}	K Pa^{-1}
Heat capacity of water	c_w	FOLLOW UP	J kg^{-1}K^{-1}
Glen's n	n	3	
Ice flow constant at the bed[a]	A_b	6.8×10^{-24}	Pa^{-n}s^{-1}
Bed elevation	z_b		m
Ice thickness	H		m
Budd friction coefficient	C_b	100	CHECK
First friction law exponent*	r	1/4	
Second friction law exponent*	s	1/4	
Channel conductivity	k_c	**0.25–0.01**	m$^{\frac{3}{2}}$kg$^{-\frac{1}{2}}$
Sheet conductivity[b]	k_s	**0.05–0.005**	Pa s^{-1}
Englacial void ratio	e_v	1×10^{-4}	
First sheet flow exponent	α_s	5/4	
Second sheet flow exponent	β_s	3/2	
First channel flow exponent	α_c	5/4	
Second channel flow exponent	β_c	3/2	
Transition parameter	ω	1/2000	
Basal bump height	h_b	**0.2–0.05**	m
Channel sheet width	l_c	**50–5**	m
Cavity spacing	l_r	**20–1**	m
Basal melt rate	M_r	**5–0.1**	m a^{-1}
Lake refill rate	Q_in	**20–5**	m^{3}s^{-1}
Elastic sheet depth scale	h_c	0	m
Elastic sheet exponent	γ	1	
Uplift regularisation rate	h_ε	1×10^{-6}	m Pa^{-1}
Regularising pressure	N_ε	1×10^4	Pa

[a]The bed is assumed to be at pressure melting point, but the ice column is assumed equivalent to -10°C or $A = 4.9\times10^{-25}$ Pa^{-n}s^{-1}

[b]Note that following Hill and others (2024) the units for

flood cycle and into the following in shown in Fig. 2g. At the start of a flood cycle (Fig. 2c), during the lake refilling phase (Fig. 2a) the system is characterised by limited channelised drainage and $\nabla\phi$ generally perpendicular to the ice surface slope. Close to the lake outlet ($X > 9$km), contours of hydraulic potential appear concave forming a 'ridge' at the lake outlet. Sheet discharge, q_s, is directed perpendicular to these contours and such ridges indicate sheet flow out of the lake. Two short (<100 m), low-discharge ($\leqslant$1 m^{3}s^{-1} channels remain open following the previous flood cycle, operating at the terminus and connected to the lake outlet.

During the lake filling phase, increasing l_h drives ϕ_outlet up (Eq. 5), resulting in a steeper $\nabla\phi$ across the domain and promoting channelised drainage close to the lake outlet [SUPP MOVIE]. Once $l_\mathrm{h} \approx 75$ m (when $t \approx 7.9$yrs) Q_r begins to rapidly exceed the lake refill rate, Q_in and an arborescent channel structure extends outwards across the full-width of the domain, perpendicular to the contours of ϕ, and therefore offset to the direction of ice-surface slope. Closer to the ice terminus, contours of ϕ become increasingly convex, and channel tributaries coalesce into three primary channels which evacuate water from the system. As the lake level drops sharply during a flood, ϕ_outlet drops, reducing $\nabla\phi$, and in turn reducing Q_c below the level necessary to sustain channels across the domain, including at the lake outlet. Once $Q_\mathrm{r} < Q_\mathrm{in}$ the flood terminates and the cycle begins anew.

Basal velocity, u_b changes significantly throughout a flood cycle, gradually accelerating during the flood initiation phase and abruptly decelerating $\approx$two months before the flood peak (Fig. 2g). Ice flow acceleration propagates from the lake outlet towards the terminus, with maximum velocity sustained within 2 km of the lake outlet.

3.2 Parameter sensitivity

Fig. 3 shows the time series from a total of 14 simulations across the maximum (red lines), minimum (blue lines) parameter values with the reference default simulation in grey. The majority of simulations yield stable and recurring asymmetric lake filling and drainage cycles (Fig. 3a[i]–f[i]) with corresponding floods in channel discharge at the lake outlet, Q_r (Fig. 3a[ii)]–f[ii]). Amongst the parameters tested, the clearest influence is on the amplitude of lake height and its rate of change, both of which in turn influence the timing of flood initiation and peak flood discharge, with higher amplitude flood cycles leading to higher peak flood discharges.

Increasing k_c, in effect making channels more sensitive to $\nabla\phi$ (see Eq. A11), leads to lower lake highstands, higher peak discharge, earlier flood onsets and total lake drainage. At maximum k_c the model becomes physically unstable—the lake drains completely each flood cycle as peak discharge and flood duration increase until $\sim$2 years into the simulation when Q_r increases rapidly and continuously. Conversely, at minimum k_c, the capacity of channels to drain the lake is inhibited and flood onset is delayed. Because channels grow slowly, peak Q_r is also dampened leading to relatively long flood cycle durations ($\sim$ 5 years).

Increasing k_s increases the capacity of the sheet to accommodate water and delays transmission to channels, this results in reduced lake highstands and delayed flood initiation as the onset of classical channel growth is delayed. Reducing k_s increases the relative efficiency of channels and leads to more vigorous lake-fill drain cycles, which shorter and more intense floods as a result. Changing h_b has a similar effect to reducing k_s, although at the maximum bump height, the sheet becomes so efficient that it can accommodate all lake discharge and within 1 year the lake reaches a stable equilibrium where $Q_\mathrm{r} = Q_\mathrm{in}$.

Changing the contributing width of the sheet to the channel (l_c) or the representative horizontal spacing of cavities (l_r) has similar influence on lake drainage by altering the relative efficiency of channelised and sheet drainage. Reducing l_r and increasing l_c reduces the amplitude of lake height and lowers peak discharge, and vice versa when increasing l_r and reducing l_c.

Finally, increasing input of water into the either the glacier via basal melt rate, M_r or lake via Q_in enhances the amplitude of l_h oscillations and in turn peak flood discharge.

In all simulations, the maximum lake height remains below the maximum ice thickness and in the majority of simulations the lake lowstand does not fall below $\sim$10 m, only draining completely when $k_\mathrm{c} = 0.25\ \mathrm{m}^{\frac{3}{2}}\mathrm{kg}^{-\frac{1}{2}}$, and when $h_\mathrm{b} = 0.05$ m.

The characteristics of the lake filling-drainage cycle across our parameter space also strongly influences the resulting basal velocity signal, and Figure 4) shows the width-averaged velocity during at least one complete flood cycle for k_c, k_s, and h_b. Each model follows the reference case with velocity peaking before peak discharge is reached, and dropping rapidly thereafter along the falling limb of a flood. In general, shorter more intense flood cycles lead to a more rapid and higher magnitude velocity response, whereas more broad and less intense flood cycles yield a similar velocity response, maintaining peak velocity for a longer duration. However, the highest basal velocity (> 300

4 MODEL APPLICATION TO A REAL DOMAIN

We run the model on Isunnguata Sermia, an outlet glacier in West Greenland draining a catchment of $\approx 16\,000$ km^2. A recently described ≈ 3 km^2, ≈ 100 m deep ice-dammed lake (Dømgaard and others, 2024; Livingstone and others, 2025) abuts the northern margin of Isunnguata Sermia's tongue (Need better way to describe this?). The lake drains subglacially with a periodicity of 1–3 years, undergoing at least 12 fill-drain cycles in the observational period between 1987–2024 (Livingstone and others, 2025). Field data, including passive seismics, GNSS, and time-lapse imagery is available for one drainage in 2023 (see; Livingstone and others, 2025). Two clear ice-contact boundaries are visible, with the lake predominately draining subglacially along a contact at its southwestern margin, and refilling by a proglacial river at its eastern margin (Livingstone and others, 2025).

We discretise our Isunnguata Sermia domain (Source of original catchment?) using an anistropic mesh with an edge length of XX m close to terminus, dropping to XX km close to the upper margin of our domain at the approximate ELA (at $H \approx 1600$ m). Bedmachine vX.X is used for z_b and H [REF]. The MAR [REF surface mass balance (SMB), runoff, and accumlation is used as climate forcing between XXXX–XXXX. An initialisation u_b is taken from the annual MEASURES mosaic 2021, from which we invert for the basal friction coefficient using an L-curve scheme (following; Wolovick and others, 2023). Because we have an observational constraint on ice-surface velocity, we use a Schoof-style friction law (see Section 2.1) which has been shown to more faithfully represent ice-dynamics with a GlaDS derived N than the Budd-style friction law used in are simple synthetic experiments. We invert for the basal friction coefficient C using an L-curve analysis to select appropriate regularisation parameters (as in; Wolovick and others, 2023). Geothermal flux is taken from REF, the Karlsson[REF] basal melt rates are used, and ice temperature is taken from REF.

Our GlaDS parameters are given in Tables 2 & A.1, with changes to the specific values intended to reflect the anticipated change in the scale of basal drainage elements relative to our synthetic domain (Hill and others, 2024). As boundary conditions, we set ϕ equivalent to atmospheric pressure at the lowest elevation vertices within a cluster adjacent to a proglacial river at the glacier terminus, and set two ice-contact boundaries at lowest elevation nodes along the southwestern and northeastern lake boundaries.

Our experimental approach was split into two phases, the first of which was to obtain a basal friction coefficient C. We ran an initial inversion for C using an empirically-derived N taking into account bed

Table 3. Parameters in our Greenland run, listing those which differ from Table 2

Variable	Units	Description

topography and ice thickness, before running a simple GlaDS-only model run over 10 years, which did not an ice-marginal lake or surface melt, to derive a more realistic N from which a second inversion for basal friction could be obtained. This approach has previously been demonstrated to yield more faithful basal inversions in a coupled hydrology model (see; Ehrenfeucht and others, 2023; McArthur and others, 2023).

Our primary model phase, in which we ran our model for a total of 30 years with a 5 minute timestep, consisted of a steady state phase, followed by a 'spin-up' into a full transient simulation. In the steady state phase, we ran our model for 15 years, defining steady state as a 95th percentile difference in sheet thickness between successive timesteps which fell below 1×10^{-6}m. During the steady state phase, we allowed the ice-marginal lake to freely evolve from an initial height of $l_\mathrm{h} = 30$ m. This steady state phase allowed the model to reach a condition which approximates wintertime hydrological conditions where there is limited channel activity and water pressures across the domain are close to the flotation fraction ($p_{\mathrm{w}\approx0.7--0.9\times p_\mathrm{i}}$). In the transient phase, taking the results of the steady-state run as initial conditions, we spun our model up over five years gradually increasing the forcing by 20% each year. Climate forcing was gradually introduced by increasing the 2013 runoff by 20% each year. Following this spin-up, the full transient simulation proceeded from 2013 onwards using the complete MAR timeseries from 2013–2024 as the climate forcing.

5 DISCUSSION

model reproduces 1) many observed characteristics of Jokulhlaup theory, 2) matches well with Felix and Jonny's work, and 3) successfully reproduces many of the characteristics of IS flooding

However it is not perfect, initial conditions are a key uncertainty, as well as the other physically uncertain GlaDS parameters, noting we only carried out a partial sweep here, others were run but had limited influence on drainage characteristics . Something akin to Tim's Gaussian GP emulation might offer a way to more rapidly test the parameter space and select an optimum starting condition.

Future work

6 CONCLUSIONS

7 AUTHOR CONTRIBUTION STATEMENT

The research was conceptualised by AJH, RB, AS, and SL. AJH developed the model, carried out the experiments, and prepared the analysis. AJH wrote the manuscript, with input from RB, AS, and SL. All authors reviewed and edited the manuscript, and fieldwork to deploy and maintain the sensors was carried out by everyone other than AJH [sad-face]. Funding was acquired by AJH, SL, RS, SB, AS, KL, KW, JG, EB, NR, LE, BG, TH, MPJ and AB.

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9 PROPER NOUNS WILL BE CORRECTED IN REFERENCES PRE-SUBMISSION...

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A ADDITIONAL MODEL DESCRIPTION

A.1 The Glacier Drainage System model

As described fully in Werder and others (2013), the hydraulic potential at the bed, ϕ, is defined at the bed by:

$$\phi = \phi_{\mathrm{m}} + p_{\mathrm{w}}, \tag{A1}$$

with water pressure, p_{w}, and elevation potential

$$\phi_{\mathrm{m}} = \rho_{\mathrm{w}} g z_{\mathrm{b}}, \tag{A2}$$

where ρ_{w} is water density, g is gravitational acceleration and z_{b} is bed elevation. In turn, effective pressure N is given by:

$$N = p_{\mathrm{i}} - p_{\mathrm{w}} = \phi_{\mathrm{O}} - \phi, \tag{A3}$$

where p_{i} is ice overburden pressure ($p_{\mathrm{i}} = \rho_{\mathrm{i}} g H$) and ϕ_{O} is the overburden potential ($\rho_{\mathrm{O}} = \phi_{\mathrm{m}} + p_{\mathrm{i}}$).

In the original GlaDS (Werder and others, 2013), sheet discharge, q_{s} is given by:

$$q_{\mathrm{s}} = -k_{\mathrm{s}} h_{\mathrm{s}}^{\alpha} |\nabla\phi|^{\beta} \nabla\phi, \tag{A4}$$

where k_{s} is the sheet conductivity; h_{s} is the sheet thickness; and α and β are flow exponents. Following the Darcy-Weisbach law, when flow is fully turbulent the flow exponents $\alpha = 5/4$ and $\beta = 3/2$. Here we use the Hill and others (2024) laminar-turbulent transition model, and with $\alpha_{\mathrm{s}} = 5/4$ as here, Eq A4 becomes:

$$\begin{aligned} q_{\mathrm{s}} = &-\frac{\nu}{2\omega}\left(\frac{h_{\mathrm{b}}}{h_{\mathrm{s}}}\right)^{1/2} \\ &\cdot \left(-1 + \sqrt{1 + 4\frac{\omega}{\nu}\left(\frac{h}{h_{\mathrm{b}}}\right)^{1/2} k_{\mathrm{s}} h^3 |\nabla\phi|}\right) \frac{\nabla\phi}{|\nabla\phi|} \end{aligned} \tag{A5}$$

where ω is a laminar–turbulent transition parameter (corresponding to the Reynolds number at which water becomes fully-turbulent), h_{b} is the bed bump height, and v is the kinematic viscosity of water at 0°C.

In GlaDS, sheet thickness refers to the thickness of the cavity layer only and there is no representation of viscoelastic deformation of the overlying ice once $p_{\mathrm{w}} > p_{\mathrm{i}}$. Here, following Stevens and others (2022, based on earlier work by Hewitt 2013), we separate the distributed sheet into elastic sheet thickness h_{el}, and cavity sheet thickness h_{cav}, adding them together to give the overall sheet thickness, $h_{\mathrm{s}} = h_{\mathrm{el}} + h_{\mathrm{cav}}$. The elastic sheet thickness is directly related to the water pressure by:

$$h_{\mathrm{el}} = h_c \left(\frac{p_{\mathrm{w}}}{p_{\mathrm{i}}}\right)^{\gamma} + h_{\varepsilon 2}\left[-N_{-} + \frac{1}{2}N_{\varepsilon}\left(1 - N_{+}/N_{\varepsilon}\right)_{+}^{2}\right], \tag{A6}$$

where h_{c} is the elastic sheet depth scale, γ, is the elastic sheet exponent, h_{ε} is the uplift regularisation rate, and N_{ε} is the regularising pressure for uplift regularisation rate. In addition, $N_{-} = \min(N, 0)$ and $N_{+} = \max(N, 0)$, with the second term designed to increase rapidly once $N < 0$, as a basic representation of hydraulic jacking.

The cavity sheet thickness evolves through time given by:

$$\frac{\partial h_{\mathrm{cav}}}{\partial t} = w - r, \tag{A7}$$

for functions w and r, which describe the cavity opening and closing rate, respectively (Kamb, 1987; Walder and Fowler, 1994). Cavities open at a rate given by basal velocity u_{b} over basal bumps with height, h_{b}:

$$w(h) = \begin{cases} u_{\mathrm{b}}(h_{\mathrm{r}} - h_{\mathrm{s}})/l_{\mathrm{r}} & \text{if } h < h_{\mathrm{r}} \\ 0 & \text{otherwise} \end{cases} \tag{A8}$$

[CHECK CODE TO ENSURE THIS IS h_s not h_{cav}] where l_{r} is the horizontal cavity spacing. Viscous ice deformation leads to cavity closure, related to the effective pressure, N by:

$$r(h, N) = Ah|N|^{n-1}N, \tag{A9}$$

where A is the rheological constant of ice, multiplied by a first order geometrical factor (h, but double check it is sheet thickness), and n is Glen's flow law exponent.

Sheet elements exhcange water with channels, the cross-sectional of which, S, evolve through time due to the dissipation of potential energy, Π, sensible heat exchange, Ξ, and cavity closure rates due to viscous ice creep $v_{\rm c}$

$$\frac{\partial S}{\partial t} = \frac{\Xi - \Pi}{\rho_{\rm i} L} - v_c, \tag{A10}$$

where $\rho_{\rm i}$ is the ice density and L is the latent heat of fusion.

Finally, channel discharge, $Q_{\rm c}$ is also related to the hydraulic potential gradient by:

$$Q_{\rm c} = -k_{\rm c} S^{\alpha_{\rm c}} \left| \frac{\partial \phi}{\partial s} \right|^{\beta_c - 2} \frac{\partial \phi}{\partial s}, \tag{A11}$$

where $k_{\rm c}$ is the channel conductivity; s is the horizontal coordinate along a channel; and $\alpha_{\rm c}/\beta_{\rm c}$ are channel flow exponents.

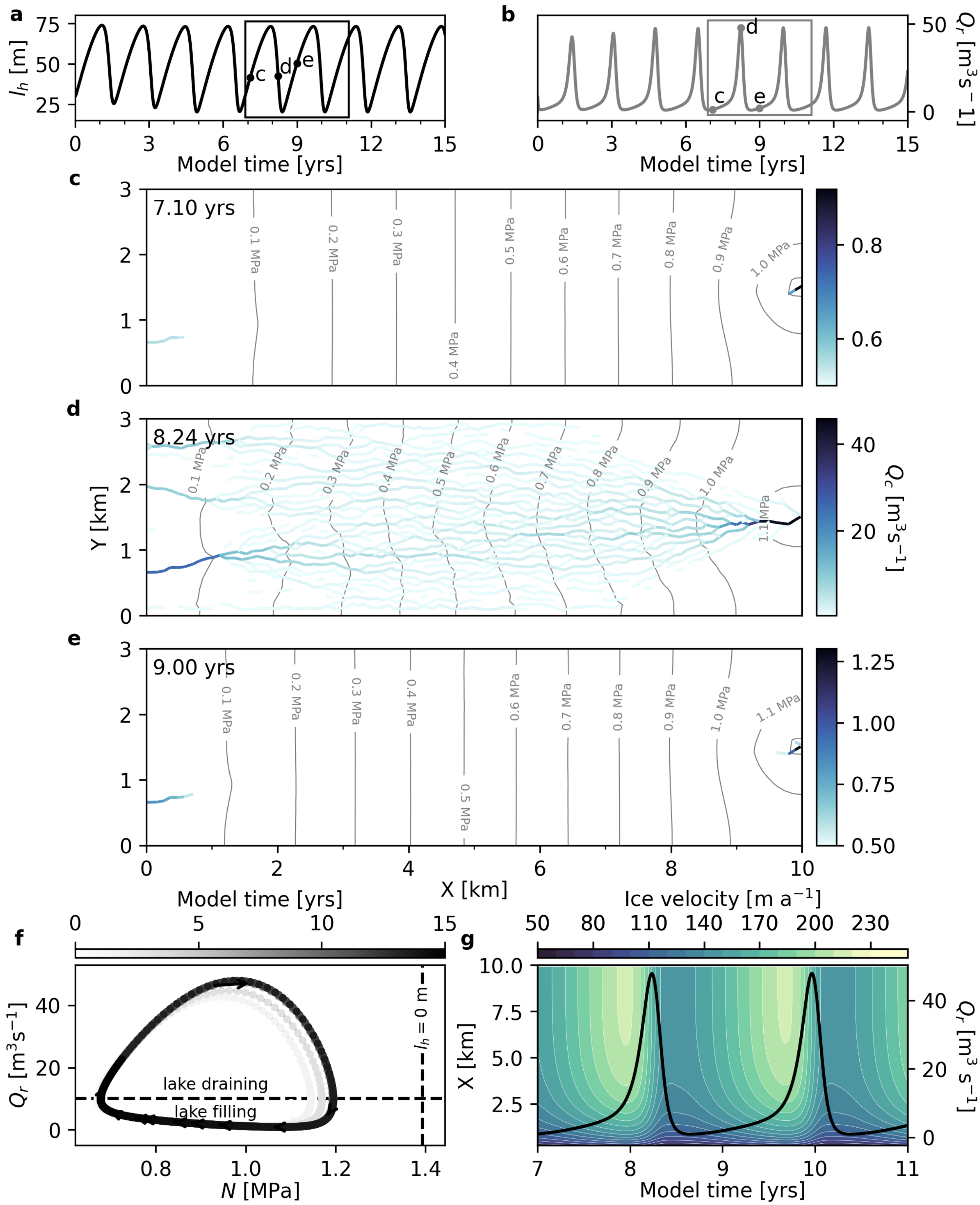

Fig. 2. [CHANGE N IN F TO BE ϕ, move channels to seperate 5 panel figure]Evolution of the 'reference' model configuration in our synthetic modelling domain. **a**) Lake height, l_{h} through time. Black points indicate l_{h} in panels **c**–**e**. **b**) Sum discharge at the lake outlet, Q_{r} through time. Grey points indicate Q_{r} in panels **c**–**e** the grey box indicates the time span of panel **g**. **c**–**e**) Channel discharge, Q_{c} at t =7.10 years (**c**), 8.38 years (**d**) and 9.00 years (**e**). In each panel, effective pressure, N is shown as grey contours. Note the changes in colour bar scale between each panel. **f**) Phase-plane diagram showing the evolution of Q_{r} and N through time at the ice-marginal lake outlet. Arrows and shading indicate the direction of time. The horizontal dashed line denotes the lake refill rate, $Q_{\mathrm{in}} = 10$ m^{3}s^{-1}. The vertical dashed line indicates N when l_{h} =0 m. **g**) Filled contour plot showing the across-flow averaged velocity through time. White contours represent velocity (every 10 ma^{-1}) and the black line shows Q_{r} between years 7–11.

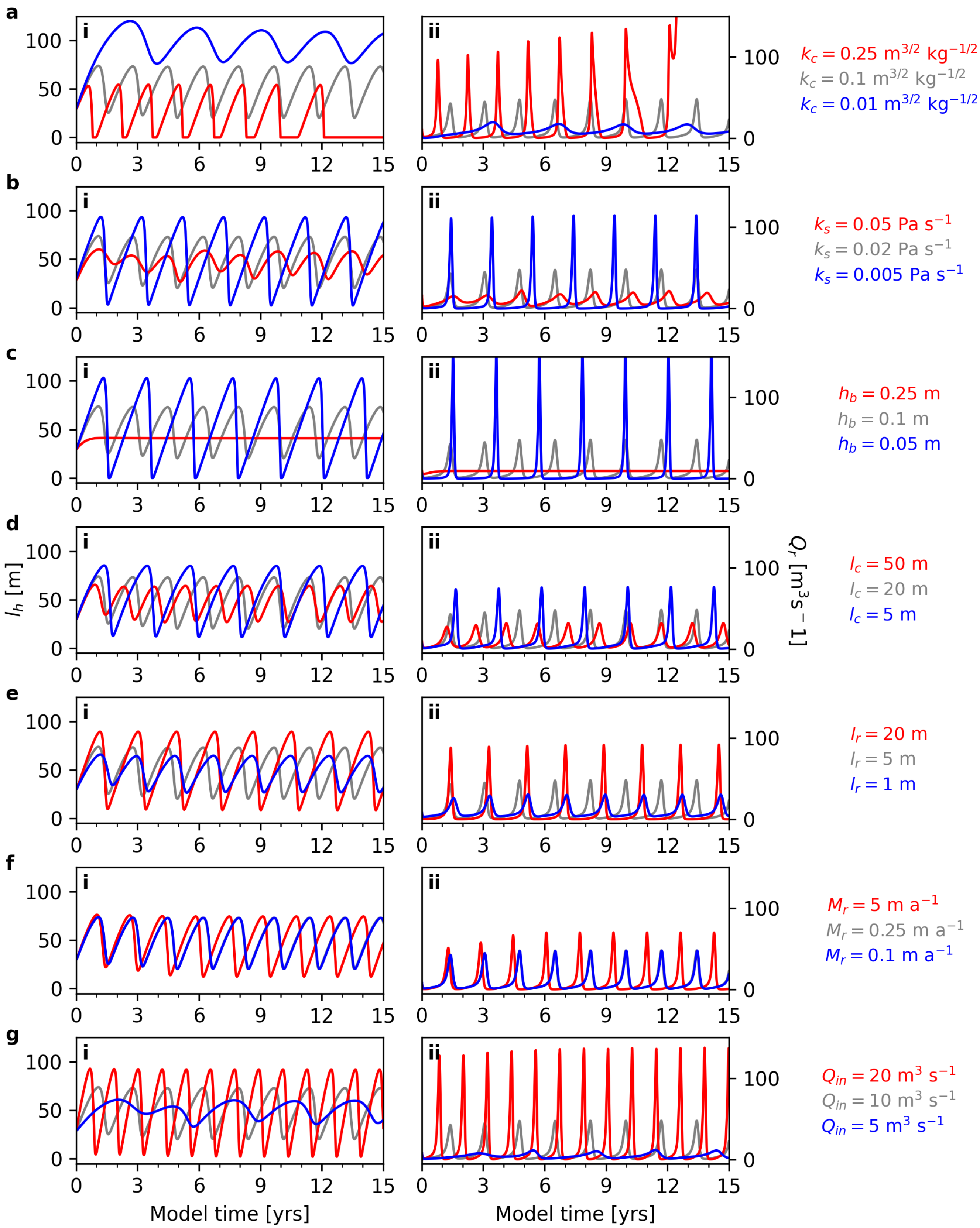

Fig. 3. NOT FINAL VERSION! Lake height, l_h (left panels, i) and sum discharge at the lake outlet, Q_r (right panels, ii) across a range of key model parameters. **a**) Channel conductivity, k_c. **b**) Sheet conductivity, k_s. **c**) Basal bump height, h_b. **d**) Channel sheet width, l_c. **e**) Cavity spacing, l_r. **f**) Basal melt rate, M_r. **g**) Lake refill rate, Q_{in}. In all panels, the default parameter run is shown in grey.

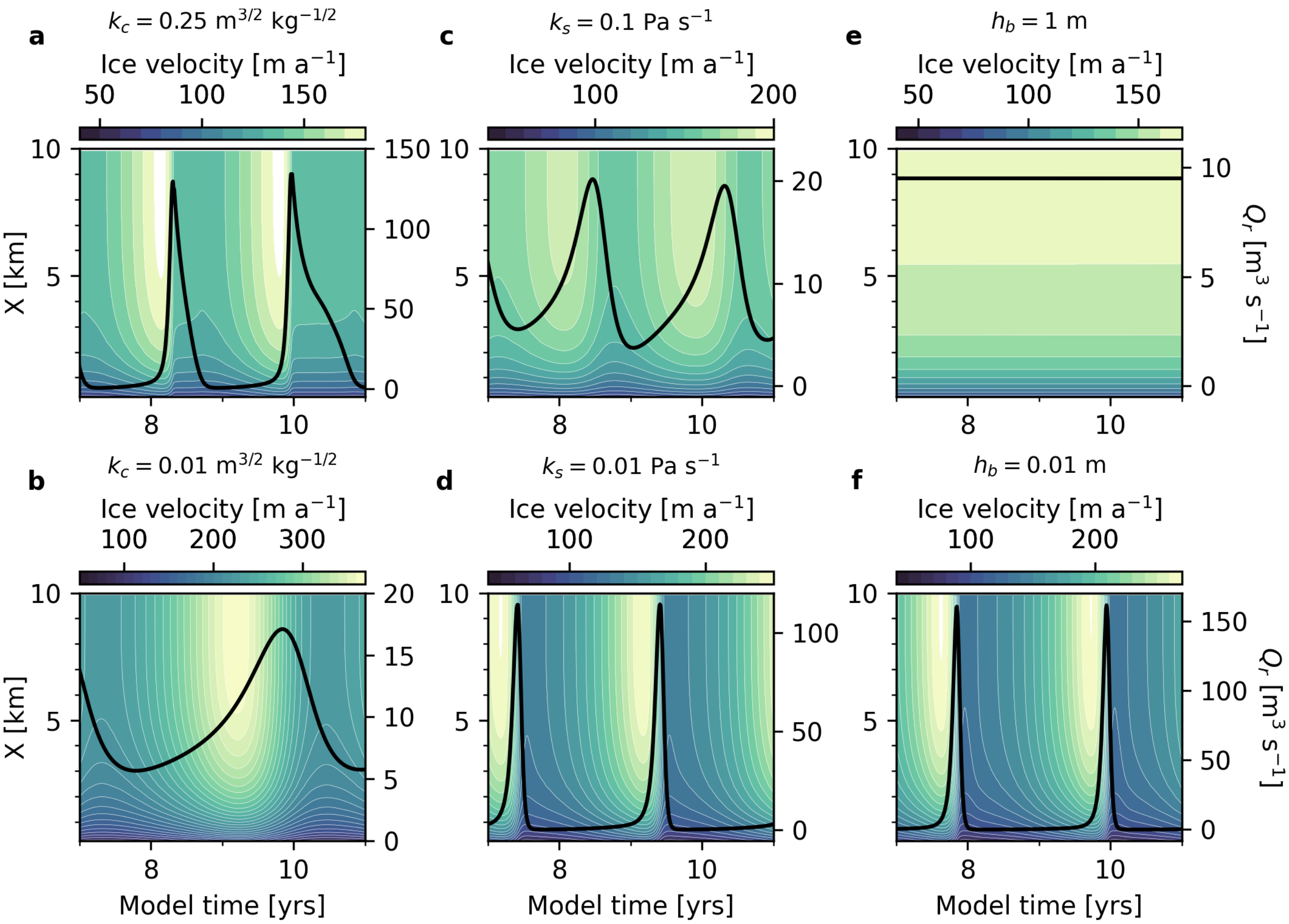

Fig. 4. NOT FINAL VERSION!